QUALITY ASSURANCE · REUSABLE TEMPLATE

Generic System QA Guide

A reusable test guide for QA assessors of any web or mobile system: access and roles, the core process flow, monitoring screens, charts, every input and output, and the checks every system shares, with numbered test cases, expected results, and sign-off.

Before you test | Access & roles | Process flow | Monitoring | Visualizations | Inputs & outputs | Cross-cutting | Templates

Applies to

Web and mobile systems with users, records, workflows, and reports

Covers

5 parts and 7 appendices, 216 numbered test cases

Fill in before testing

System name, build, environment, roles, and scope (System profile)

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BEFORE YOU TEST

How to use this guide

This guide is a starting point for QA assessors testing any system that has user accounts, records, workflows, and reports. It follows the order of a system-specific QA guide: prepare the environment and accounts, check access, walk the core process from start to finish, review each role's monitoring screens, then check charts, inputs, outputs, and the behavior shared across the whole system. Every part lists numbered test cases with expected results and a blank Result column.

1. Tailoring the guide to your system

1. Fill in the **System profile** below: name, build, environment, roles, and the units a user's data is limited to.
2. Replace each green placeholder, such as <record>, <unit>, and <approver role>, with your system's own term.
3. Copy the exact statuses, limits, and messages into Appendices A1 and A2 before testing, so expected results can be checked word for word.
4. Mark a case that does not apply as **N/A** with the reason. Do not delete it, so totals stay comparable between test rounds.
5. Add system-specific cases with the stage template (Part A) and the chart catalog (Part C). Give each the next free ID in its part.

PLACEHOLDERS USED IN THIS GUIDE

PLACEHOLDER	REPLACE IT WITH
<record>	The main thing the system manages: an order, request, application, ticket, booking, case, or parcel.
<unit>	The boundary a user's data is limited to: a branch, office, municipality, department, project, or tenant.
<approver role>	The role that reviews and approves a <record>. Use each role name exactly as the system shows it.
<balance>	Any quantity the process moves: stock, budget, credits, seats, hours, or money.

2. System profile

Fill this in once per test round. Never write passwords here.

FIELD	VALUE
System name	
Version / build	
Test environment URL	
Database or data copy used	
Test period	
Roles	
Scope units	

FIELD	VALUE
Core process (one line)	
External services (email, maps, payments, SMS, AI)	
Supported browsers and devices	
Reference documents (requirements, user manual)	

3. Scope and structure

PART	WHAT IT COVERS	TEST CASE PREFIX
Before you test	Environment, accounts, result codes, severity, entry and exit criteria, and access and role checks.	AC
A. Process flow	The main <record> lifecycle: prerequisites → create → edit → submit → review → fulfil → confirm → reverse → archive, then end-to-end reconciliation.	PF0–PF8, RC
B. Monitoring	Dashboards, headline figures, filters, lists, search, sorting, drill-down, maps, and live updates.	MO
C. Visualizations	Behaviors every chart must share, a guide to the right chart form, and a chart catalog to fill in.	VZ
D. Inputs and outputs	Input validation by field type, data outputs and where to verify them, notifications, emails, reports, and exports.	IN, OU, RP
E. Cross-cutting	Dialogs and feedback, responsive layout, accessibility, browsers, performance, network and offline, security, data integrity, time and locale.	XC, RL, AX, BC, PE, NW, SE, DI, TL
Appendices	Status map, business rules, boundary values, documentation differences, defect report, summary and sign-off, glossary.	—

4. Recording results

Each test case has a blank Result column. Write one of the codes below, and for anything other than Pass, raise a defect with the template in Appendix A5.

CODE	MEANING
P Pass	The expected result happened exactly as written, including messages, statuses, and every updated figure named in the case.
F Fail	Any part of the expected result did not happen, or something extra happened: a wrong figure, an extra record, or a record outside the user's scope.
B Blocked	The case could not run because an earlier step failed or a prerequisite (account, data, external service) was missing. Note the blocking case ID.
N/A	The case does not apply to this system, build, or environment. Write the reason.

SEVERITY

SEVERITY	USE WHEN	EXAMPLES
Critical	Data is wrong or lost, totals do not reconcile, there is a security hole, or a user sees or changes records outside their role or <unit>.	A payment is recorded twice; a Branch B manager approves a Branch A request; a deactivated account can still sign in.
High	A core workflow cannot be completed and there is no workaround.	Submit fails for valid input; a report never finishes downloading.
Medium	A wrong display, filter, or message, with a workaround.	A chart ignores the date filter; an error message names the wrong field.
Low	Cosmetic or wording issues.	A misaligned button; inconsistent capitalization; a typo.

Severity is the harm a defect does and is set by QA. Priority, how soon it is fixed, is set by the project lead and usually follows severity.

5. Test environment and data

Never test against live data

Run this guide on a staging or test environment with its own database (a copy, a branch, or a seeded instance). Test records created in production distort reports, can send real emails or payments, and are hard to remove completely.

- **Browsers and devices.** A current Chrome or Edge on desktop, plus one Android and one iOS phone if the system is used away from a desk. Record versions in every defect.
- **One browser profile per role.** Open each role in its own browser profile or private window so you can switch roles without signing out.
- **Known starting data.** Write down counts and <balance> values before you start, so every change you make can be traced. Where possible, create test records in an empty period, such as a new year.
- **Email and messages.** Use inboxes you control or a mail-catcher service. Never use real customer or staff addresses.
- **External services.** Note which services (maps, payments, SMS, AI, routing) are real, sandboxed, or mocked, and any working hours, quotas, or rate limits.
- **Dates and time zone.** Note the system's time zone. Know which actions depend on today's date, and test near midnight and at month and year ends (Part E, TL).
- **Clean-up.** Keep a list of what you created. Delete the test environment or its data after sign-off.

6. Accounts to prepare

Use one <unit> (Unit A) for the main flow and a second (Unit B) for scope checks. Give test accounts names that make test data obvious, such as QA Tester 01.

ACCOUNT	PURPOSE	HOW TO PREPARE
System administrator	Provisions accounts, roles, and settings.	Created by the technical team; never a personal account.
<unit> administrator (Unit A)	Approves and manages Unit A's users and <record>s.	Provisioned or approved through the normal flow.
<unit> administrator (Unit B)	Scope checks: must never see or change Unit A data.	Same as Unit A, for a second unit.
Standard user (Unit A) × 2	Everyday data entry. Two users test ownership, hand-over, and simultaneous edits.	Self-registered and approved, or provisioned.
Read-only viewer	Monitoring and reports only.	Provisioned with the viewer role.
Pending account	Registered but not yet approved.	Register and leave unapproved.
Blocked account	Deactivated, restricted, or locked by an administrator.	Approve, then block it through the admin tools.

SEED THE MINIMUM DATA

- Reference lists the core flow needs (categories, types, locations, price lists) loaded and active.
- At least one <record> in each status of the status map (Appendix A1).
- One <record> with the longest allowed values in every text field, to test layouts.
- Records in at least two <unit>s, so scope can be checked in both directions.
- History for earlier periods if dashboards compare periods (this year against last year).

7. Roles, screens, and scope

Copy each role's menu or sidebar names exactly as built. This table becomes the role matrix the access, process, and monitoring cases refer to.

ROLE	SCOPE	SCREENS OR SECTIONS (AS BUILT)	CAN CREATE / APPROVE / EXPORT

8. Entry and exit criteria

START TESTING WHEN	FINISH TESTING WHEN
• The build is deployed to the test environment and its version is recorded. • A smoke test passes: every role can sign in and open its home screen. • Accounts and seed data are ready; the System profile is filled in. • Requirements or the user manual, and the known-issues list, are available.	• Every case has a result (P, F, B, or N/A with a reason). • No open Critical or High defects; Medium defects have an agreed fix plan. • Fixed defects are retested, and affected cases are rerun as a regression check. • The test summary is signed in Appendix A6.

9. Access and role checks

Run these before the process flow. Every protected screen and action should check, on the server, that the user exists, has the right role, is approved and active, and is inside the right <unit>.

ID	TEST STEPS	EXPECTED RESULT	RESULT
AC-01	Sign in with each approved account.	Each account lands on its own home screen or dashboard; the header shows the right name and role.	
AC-02	Sign in with a wrong password, then with an email that has no account.	Same generic message for both; it does not reveal whether the email exists.	
AC-03	Fail to sign in repeatedly (the documented limit plus one).	Further attempts are slowed, locked, or challenged as documented; the account owner can recover.	
AC-04	Register a new account through self-registration, confirm the email if required, then sign in before approval.	Sign-in is refused with a pending-approval message that says who approves; no screen with data opens.	
AC-05	As the <approver role>, approve the pending account; sign in again with it.	The account opens its home screen. The audit log records the approval with the approver and time.	
AC-06	Reject a second pending registration.	Sign-in shows a rejected message; the account cannot open any screen with data.	
AC-07	Self-registration form: check which roles can be chosen; then try to send a privileged role by editing the request.	Only self-service roles are offered. A privileged role sent directly is refused by the server.	
AC-08	Signed in as one role, type the address of another role's screens.	Refused or redirected; no other role's data is shown, not even briefly while the page loads.	
AC-09	Signed in as the Unit B administrator, look for Unit A's users, <record>s, and figures in lists, search, counts, and exports.	None appear; every list, count, and file is limited to Unit B.	
AC-10	Open a <record> you may see, then change its ID in the address bar or request to one you may not see.	Refused (for example 403 or 404); no data from the other record is shown.	
AC-11	Forgot password: request a reset, open the email on a different device, and set a new password.	Reset works on the other device. Using the same link again says it was already used or has expired.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
AC-12	Set a password that breaks each password rule (length, common password, mismatch).	Each rule is shown and enforced; the new password works only when all rules pass.	
AC-13	Leave a signed-in session idle past the documented timeout, then act.	Asked to sign in again; the action is not completed silently. Unsaved input is kept or the loss is warned.	
AC-14	Sign out on a shared device, then press Back.	No data is shown until someone signs in again.	
AC-15	Block (deactivate or restrict) an account while it is signed in elsewhere.	Its next action or page load is refused and it cannot sign in again; the block is logged with a reason.	
AC-16	Reactivate the account from the previous case.	It returns to its earlier status and can sign in. A pending account that was blocked returns to pending, not approved.	
AC-17	Change a user's role, then use the account.	New permissions apply from the documented moment (next action or next sign-in); old permissions are gone.	
AC-18	Open an invitation or account-setup link twice, and once after it expires.	Works once; afterwards it says the link was used or has expired.	
AC-19	Perform an action the role may not do by replaying the request (browser developer tools or an API client).	The server refuses it; nothing changes. Hiding a button is not the only protection.	
AC-20	Sign in to the same account on two devices.	Allowed or refused as documented; signing out on one behaves as documented on the other.	

PART A · PROCESS FLOW

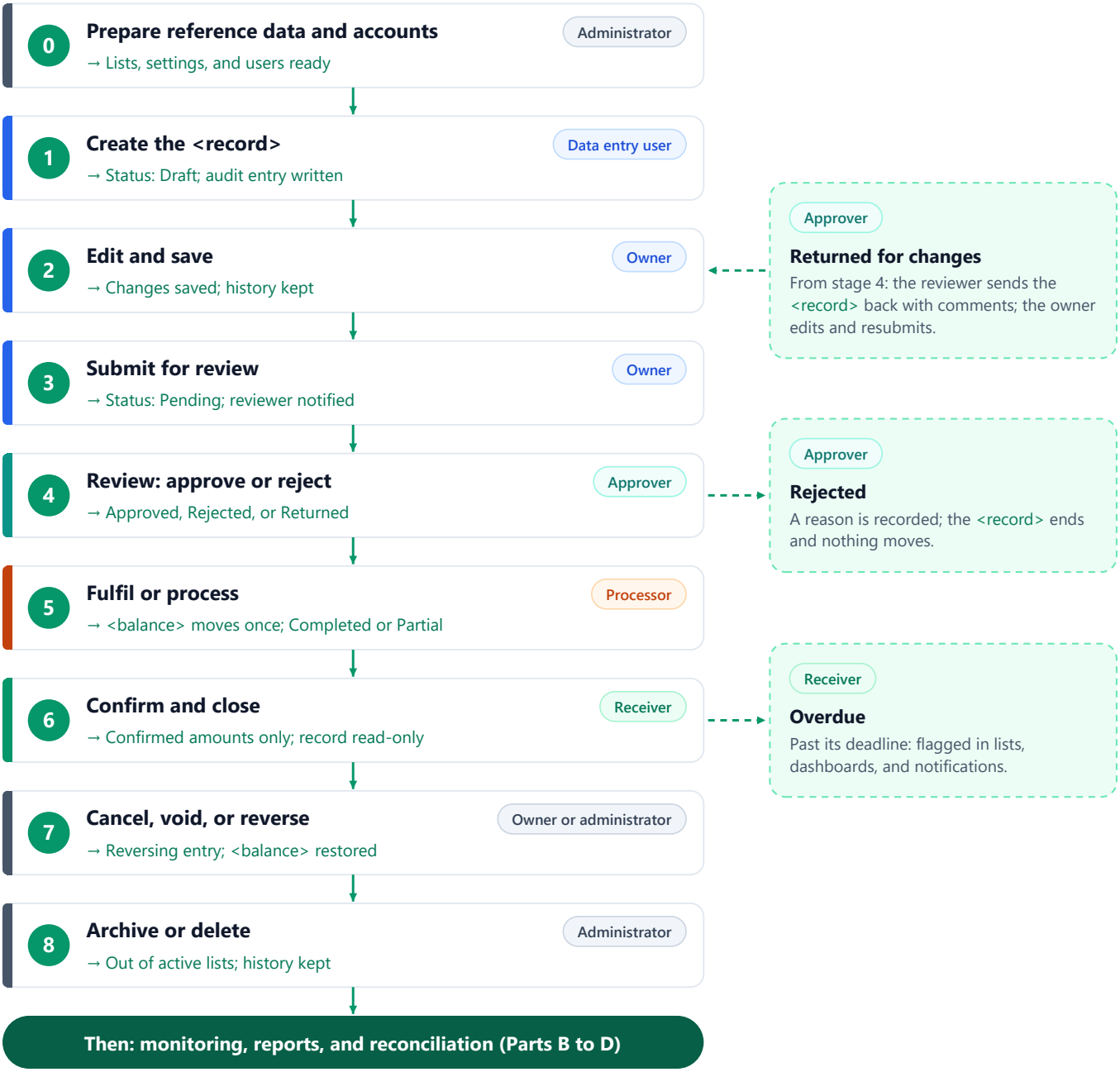
The core process, stage by stage

Most systems move a main <record> through a lifecycle: someone creates it, others review and act on it, and it ends confirmed, cancelled, or archived, moving a <balance> along the way. Map your system's flow onto the stages below, then run each stage's cases on the real screens.

Stage index

STAGE	WHAT HAPPENS	TYPICAL ACTOR	PREFIX
0	Prerequisites: reference data, settings, accounts	Administrator	PF0
1	Create a <record>	Data entry user	PF1
2	Edit and save a draft	Owner	PF2
3	Submit for review	Owner	PF3
4	Review: approve, reject, or return	Approver	PF4
5	Fulfil or process (moves the <balance>)	Processor	PF5
6	Confirm and close	Receiver	PF6
7	Cancel, void, or reverse	Owner or administrator	PF7
8	Archive or delete	Administrator	PF8
—	End-to-end reconciliation	QA assessor	RC

Process flow diagram



Solid boxes are the main chain; dashed boxes are branches. A <balance> should move only at the stages your system documents (usually 5, 6, and 7); every other stage only changes a status. Stage 7 can start from several stages, so test it from each one your system allows.

Stage template: documenting your own stages

A system-specific guide gives every stage a card, one or two sentences, and four tables. Copy this layout for each stage of your real flow. The generic stages that follow already include test cases you can keep, edit, or mark N/A.

N

<Stage name>

<Role>

Where: <Dashboard> › <Section> › <Tab> (and the tabs on that screen)

<One or two sentences: what the stage does, who can do it, and what it changes.>

INPUTS

FIELD	TYPE / OPTIONS	RULES AND VALIDATION
<Field name>	Text, number, date, select, or file; the options offered	Required or optional; limits and formats; the exact error message
<Second field>	…	…

CONTROLS

CONTROL	WHAT IT DOES AND WHAT TO CHECK
<Button, filter, or tab>	What happens on click; the loading text; when it is disabled; the confirm dialog; keyboard shortcuts

OUTPUTS

OUTPUT	WHERE TO VERIFY IT
<Record, status, or balance>	<Screen> › <tab>: the fields and figures that must change, and the audit entry

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PFn-01	<The action, with valid input>	<The exact result: message, status, and every figure that changes>	
PFn-02	<The same action with invalid or boundary input>	<Refused; the exact message; nothing changes>	

0 Prerequisites: reference data and settings

Administrator

Where: <Admin area> › Reference lists, Settings, Users

Everything the core flow needs before the first <record> can be created: reference lists, system settings, and accounts. Only administrators should be able to change them.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF0-01	Open each reference list the core flow uses (categories, types, locations, prices).	Every required entry exists and is active; inactive entries are not offered in forms.	
PF0-02	As an administrator, add a new reference entry; then add the same entry again.	The first is saved and appears in the related form without a code change; the duplicate is refused with a clear message.	
PF0-03	Deactivate a reference entry that existing <record>s use.	Existing <record>s still show it; new <record>s cannot select it.	
PF0-04	As a non-administrator, try to change reference data or settings (screen and direct request).	The control is hidden and the server refuses the change.	
PF0-05	Change a setting the flow depends on (a limit, deadline, or rate).	The new value applies as documented (new <record>s only, or immediately); the change is logged with the old and new value.	

1 Create a <record>

Data entry user

Where: <Module> › <Record list> › New <record>

A user enters a new <record>. Check required fields, duplicates, double submissions, and where the new <record> appears afterwards.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF1-01	Create a <record> with every required field valid.	Saved with a confirmation message; it appears in its list with the starting status, the creator, and the time.	
PF1-02	Try to save with each required field empty in turn.	Save is blocked; the message names the field; the other entries are kept.	
PF1-03	Create a duplicate (the same unique key or reference number).	Refused or flagged exactly as documented; no second record is created.	
PF1-04	Double-click Save, or press Enter twice quickly.	Exactly one <record> is created; the button is disabled while saving.	
PF1-05	Fill in the form, then press Cancel or leave the page.	Asked to confirm discarding the changes; nothing is saved.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF1-06	Create a <record> with all optional fields empty.	Saved; empty fields show as "—" or blank, never "null", "undefined", or "NaN".	
PF1-07	Create a <record> with maximum-length and special-character values (see Part D).	Saved and displayed exactly as typed; the layout does not break.	
PF1-08	Create a <record> as a Unit B user.	It belongs to Unit B; Unit A users cannot find it anywhere.	
PF1-09	Attach a file or photo, if the form allows it.	Uploaded, previewable, and linked to the <record>; the limits in Part D apply.	
PF1-10	Open the audit log.	One create entry with the actor, time, and key values.	

2 Edit and save a draft

Owner

Where: <Record list> › <record> › Edit

The owner changes a <record> before it is submitted. Check that edits persist, that locked <record>s stay locked, and that simultaneous edits never silently overwrite each other.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF2-01	Edit a draft and save; reload the page.	Changes are kept; the modified time and editor are updated.	
PF2-02	Two users open the same <record>; both change it and save, one after the other.	The second save is warned or merged as documented; the first user's change is never lost silently.	
PF2-03	Try to edit a <record> in a locked status (submitted, approved, closed), on screen and by a direct request.	Edit controls are hidden or disabled; the server refuses the change.	
PF2-04	Edit a <record> you do not own, in the same <unit>.	Allowed or refused exactly as the role matrix says.	
PF2-05	Clear a required field and save.	Blocked, as in Stage 1.	
PF2-06	Change a field that drives a calculation (quantity, area, price, rate).	Totals and other derived values update on screen and after saving.	
PF2-07	Open the change history, if the system keeps one.	Each edit shows the old and new values, the editor, and the time.	

3 Submit for review

Owner

Where: <record> › Submit

The owner sends the <record> to the <approver role>. From here the owner usually cannot edit it, and the reviewer's queue should change.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF3-01	Submit a complete <record>.	The status changes to the submitted status; the <record> becomes read-only for the owner; the reviewer is notified.	
PF3-02	Submit a <record> missing a field that is only required at submission.	Blocked, with the list of missing items.	
PF3-03	Submit a <record> that breaks a business rule (over a limit, outside a date window, not enough <balance>).	Blocked or flagged with the rule named, exactly as documented.	
PF3-04	Withdraw or recall a submitted <record>, if allowed.	It returns to draft and leaves the reviewer's queue.	
PF3-05	Check the reviewer's queue and badge counts.	The <record> appears once; counts go up by exactly one.	

4 Review: approve, reject, or return

Approver

Where: <Approvals> › Pending <record>s

The <approver role> decides. Check that only the right role in the right <unit> can decide, that reasons are required where documented, and that a decision is made only once.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF4-01	Approve a pending <record>.	The status changes to approved; the approver and time are recorded; the owner and the next actor are notified.	
PF4-02	Reject a <record> without a reason.	Blocked; a reason is required.	
PF4-03	Reject a <record> with a reason.	The status changes to rejected; the owner sees the reason; the <record> cannot continue.	
PF4-04	Return a <record> for changes, if supported.	It goes back to the owner, editable, with the comments; resubmitting sends it to the same reviewer.	
PF4-05	As the owner, try to approve your own <record>.	Not possible, unless the role matrix explicitly allows it.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF4-06	As the Unit B approver, open or approve a Unit A <record> (screen and direct request).	Refused; no data is shown.	
PF4-07	Two approvers act on the same <record> at the same time.	Only the first decision applies; the second approver is told it was already decided.	
PF4-08	Approve a <record> after something it depends on has changed (for example, the <balance> fell).	Warned or blocked as documented.	
PF4-09	Multi-level approval, if any: try to approve out of order.	Each level approves in order; a level cannot be skipped.	

5 Fulfil or process

Processor

Where: <Fulfilment> › Approved <record>s

The approved <record> is carried out and the <balance> moves. This is where double counting and lost updates usually hide.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF5-01	Process an approved <record> in full.	The <balance> moves once, by exactly the amount; the status changes to completed.	
PF5-02	Process part of the quantity.	The remainder stays open and visible; the status shows partial completion.	
PF5-03	Process more than was approved or is available.	Refused with the limit named; nothing moves.	
PF5-04	Process the same <record> twice (refresh, Back, double-click).	The second attempt is refused; the <balance> is unchanged.	
PF5-05	Process a zero or negative quantity.	Refused.	
PF5-06	Turn the network off in the middle of processing, then reconnect.	Either the whole step applied or none of it; no half-updated <balance>.	
PF5-07	Check every screen that shows an affected <balance>.	Before + change = after, in every place it is shown.	

6 Confirm and close

Receiver

Where: <record> › Confirm receipt / Close

The receiver confirms the outcome and the <record> is closed. Only confirmed amounts should count.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF6-01	Confirm with the actual quantity or outcome, including a shortfall.	Only the confirmed amount is credited; the shortfall stays open or is recorded exactly as documented.	
PF6-02	Confirm an outcome that differs from the plan without a note.	A note is required.	
PF6-03	Close the <record>.	It becomes read-only and appears in closed or history lists and in reports.	
PF6-04	Reopen a closed <record>, if allowed.	Only the permitted role can; the reason is logged; the <balance> adjusts correctly.	
PF6-05	Leave a <record> past its deadline.	Flagged as overdue in lists, dashboards, and notifications.	

7 Cancel, void, or reverse

Owner or administrator

Where: <record> › Cancel / Void / Reverse

A mistake is undone. A finished <record> should be reversed by a new entry, not erased, so the history stays complete.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF7-01	Cancel a draft or pending <record>.	The status changes to cancelled; nothing moves; the reason is recorded if required.	
PF7-02	Void or reverse a completed <record>.	A reversing entry is created and the original is kept; the <balance> returns to its earlier value.	
PF7-03	Reverse the same <record> a second time.	Refused.	
PF7-04	Void a <record> after a later step has used it.	Blocked or cascaded as documented; no orphan records remain.	
PF7-05	Open the period's report.	Both the original and the reversal appear, netting to zero.	

8 Archive or delete

Administrator

Where: <Admin area> › <record> › Archive / Delete

Records leave the active lists. Check that dependent data is protected and that deleted records are treated the same way everywhere.

TEST CASES

ID	TEST STEPS	EXPECTED RESULT	RESULT
PF8-01	Delete a <record> that nothing depends on.	It leaves the active lists (soft-deleted if documented); the deletion is logged.	
PF8-02	Delete a <record> that other records depend on.	Refused with the reason, or archived instead, as documented.	
PF8-03	Restore a deleted or archived <record>, if supported.	It comes back with its history.	
PF8-04	As a non-administrator, try to delete.	Not allowed on screen or by a direct request.	
PF8-05	Search, open dashboards, and run reports after the deletion.	The <record> is excluded or labelled the same way everywhere.	

End-to-end reconciliation

After the stages, check that the whole flow adds up. These cases catch double counting, figures that disagree between screens, and <balance>s that move at the wrong stage.

ID	TEST STEPS	EXPECTED RESULT	RESULT
RC-01	Take one <record> through every stage, writing down each <balance> after each step.	Each <balance> changes only at the documented stages; the final totals match a hand calculation.	
RC-02	Compare the same figure on the dashboard, the list, a report, and an export.	Identical for the same filters.	
RC-03	Add up the period's ledger or transaction list.	Equals the closing <balance> minus the opening <balance>.	
RC-04	Read the <record>'s status history.	Every change is allowed by the status map (A1); no skipped or backward steps.	
RC-05	Check what each role sees for the <record>.	Matches the role matrix; no role sees another <unit>'s figures.	
RC-06	Read the audit log for the whole flow.	One entry per action, in order, with the right actor.	
RC-07	Check the notifications for the whole flow.	Each actor received exactly the notifications for their steps, once.	
RC-08	Repeat the first case in a second <unit> at the same time.	The two <unit>s' totals stay independent.	

PART B · MONITORING

Dashboards, figures, and lists

Dashboards and lists turn records into figures. Every figure must be traceable to the records behind it, follow the filters it claims to follow, and stay inside the user's scope. Run these cases on each role's monitoring screens and record which screen each result came from.

Figures and filters

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-01	For each headline figure, count or add up the matching records by hand or with a query.	The figure equals the count or sum; its unit is shown.	
MO-02	Change each filter in turn (period, year, <unit>, status, category).	Every figure, chart, and list that claims to follow the filter changes. Those that should not (live queues, current <balance>) stay the same and say so.	
MO-03	Choose a period with no records.	Figures show "—" or "No records", not 0, unless 0 is a real measured value.	
MO-04	Check records dated exactly on period boundaries (first and last day, midnight, year end).	Each is counted in the correct period, in the system's time zone.	
MO-05	Compare a role's figures with a higher-level role's figures for the same <unit>.	The lower role's figures equal the higher role's figures filtered to that <unit>.	
MO-06	Check every percentage and rate.	The numerator and denominator are stated or traceable; no division by zero; rounding is consistent; shares add up to 100% (±0.1).	
MO-07	Check period comparisons (this year against last year).	The compared period is named; arrows and colors match the meaning (up is not always good).	
MO-08	Check target or threshold statuses at, just below, and just above each threshold.	The status (on track, at risk, behind) matches the documented thresholds.	
MO-09	Switch tabs and come back.	Filters are kept or reset as documented, the same way across sections.	
MO-10	Reset or clear the filters.	Returns to the documented default view, for example the current year and every <unit> in scope.	
MO-11	Check each "needs attention" or queue row.	Its count matches the list it opens; empty rows are hidden or labelled.	

Lists, search, and sorting

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-12	Page through a long list with Next and Previous, including rows that share the same date.	No row repeats or goes missing across pages.	
MO-13	Sort by each sortable column, ascending and descending.	Correct order, including blanks, numbers (10 after 9), and dates.	
MO-14	Search with partial text, different letter case, accents, and extra spaces.	Expected matches are found; a clear "No results" state appears when there are none.	
MO-15	Search with special characters: % _ ' " < &	No error; the characters are treated as text.	
MO-16	Combine search with filters, then remove one.	Results reflect every active criterion; the total count updates.	
MO-17	Check the list counts, such as "Showing 1–20 of 135".	Accurate on every page and after filtering.	
MO-18	View a cell with very long text.	Wrapped, or truncated with a way to read it in full; the table layout stays intact.	

Drill-down, live updates, and maps

ID	TEST STEPS	EXPECTED RESULT	RESULT
MO-19	Click a figure, chart item, or attention row that links somewhere.	Opens the matching records with the same filters applied.	
MO-20	Open a <record> from a list, then go Back.	Returns to the same page, scroll position, and filters.	
MO-21	Change data as another user while the dashboard is open.	The dashboard updates live or on refresh within the documented time.	
MO-22	Check cached or summary figures.	The last-updated time is shown; stale data is labelled.	
MO-23	Load a realistic year of records.	The dashboard loads within the agreed time; no timeouts or cut-off totals.	
MO-24	Maps, if any: compare markers or shapes with the records; apply filters; open popups; view on a phone.	Markers match the records and filters; popups show the right details; the map fits the screen.	

PART C · VISUALIZATIONS

Charts, figures, and maps

Charts must be accurate, readable, and consistent across the system. Run the universal checks once per screen that has charts, check that each chart uses a form that fits its data, then record every chart in the catalog.

Universal chart checks

ID	TEST STEPS	EXPECTED RESULT	RESULT
VZ-01	Hover, keyboard-focus (Tab), and tap each bar, point, slice, or cell.	A tooltip shows the exact value, unit, and label; tapping elsewhere closes it.	
VZ-02	Try to read every value without hovering.	Values are available as labels or a table view; no value exists only in a tooltip.	
VZ-03	Compare chart totals with the list or table behind them.	Equal for the same filters.	
VZ-04	Choose an empty period or filter.	An empty-state message appears; no empty axes or misleading zero bars.	
VZ-05	Find a figure based on very few records.	Labelled as too few records instead of a precise-looking percentage.	
VZ-06	Compare colors across charts and screens.	Each color means the same thing everywhere (for example green done, amber pending, red failed), and a category keeps its color.	
VZ-07	Find a chart with many categories.	The top items plus "Other"; no color repeats within one chart; the legend is readable.	
VZ-08	Find a chart that mixes two units (count and money, bags and tonnes).	Shown as separate panels or charts, never on two y-axes.	
VZ-09	Check the axes.	Bars start at zero; labels, units, and periods are present; labels do not overlap.	
VZ-10	Check time series.	Periods are continuous and in order; missing periods show as gaps, not lines drawn across them.	
VZ-11	Check part-to-whole charts.	Segments add up to 100% or the stated total; pie or donut charts are used only for a few parts of one whole.	
VZ-12	View the charts in grayscale or with a color-blindness simulator.	Meaning is also given by labels, positions, or text, not by color alone.	
VZ-13	Click a chart that filters the screen, then click the same item again.	The first click sets the filter it says it sets; the second clears it; a hint explains this.	
VZ-14	Leave a chart open while the data changes.	It updates without flicker or replaying long animations.	
VZ-15	Open the charts at phone width.	Charts fit the width; labels stay readable; wide tables scroll sideways.	
VZ-16	Print or export a screen with charts.	The charts appear in the output with the same values.	

Is it the right chart?

A chart can be accurate and still mislead if its form does not fit the data. Compare each chart with this table and raise a Medium defect when the form makes the reader misjudge the data.

WHAT THE DATA MUST SHOW	EXPECTED FORM
One headline number, with change from last period	A number tile with the change and the compared period
One measure ranked across many categories	Horizontal bars, sorted
A few categories, or periods as categories	Columns
Change over time	A line; columns for monthly totals of seasonal events
Parts of one whole, six or fewer parts	A donut, or a single stacked bar
Mix per row when rows differ in size	100% stacked bars
Two values per row (this year and last year)	A dumbbell or grouped bars
Progress toward a target	A progress bar or gauge with the target marked
Spread of values	A histogram
Stages that only lose items (applied → approved → completed)	A funnel
Two categories against each other (month × year)	A heatmap

Two numbers (this year against last year) are clearer as two figures with a change than as a two-bar chart. A share that is the same everywhere is a headline figure, not a chart.

Chart catalog

List every chart once, with where it appears, so each one is checked against the universal cases.

CHART	WHERE (SCREEN › TAB)	WHAT IT SHOWS	FILTERS IT FOLLOWS	RESULT

PART D · INPUTS AND OUTPUTS

Every user input and data output

Every input must accept what it should and refuse everything else, on the server as well as on screen. Every output (records, logs, notifications, emails, files) must say the same thing as the screens. Use the matrix to choose test values for each field type, then run the cases.

Input validation by field type

FIELD TYPE	TRY THESE VALUES	EXPECTED
Short text	Empty; spaces only; 1 character; the maximum; maximum + 1; leading and trailing spaces; accents (ñ, é); emoji; `<script>alert(1)</script>`; `' OR 1=1 --`	Required rules enforced; spaces trimmed; stored and shown exactly as text; never run as code; the length limit is stated.
Long text / notes	5,000+ characters; line breaks; text pasted from a word processor	Limit enforced with a counter or message; line breaks kept; pasted formatting removed safely.
Email	Valid; missing @; spaces; capitals; plus-addressing (name+qa@example.com); an email already in use	Invalid formats refused; uniqueness ignores letter case; plus-addresses accepted.
Phone	Local and international formats; letters; too short; too long	Accepted formats documented; stored in one consistent format.
Whole number	0; 1; −1; the maximum; maximum + 1; a decimal; leading zeros; letters; empty	Range enforced; decimals refused where whole numbers are required; no scroll-wheel changes by accident.
Decimal, money, percent	Many decimals; comma and dot separators; negative; 100.01%	Rounded by the documented rule; separators follow the locale; range enforced.
Date	Today; past; future; 29 February in a leap and a non-leap year; 31 April; typed and picked; year 1900 and 2100	Invalid dates refused; past and future rules enforced; shown in the local format.
Date range	End before start; the same day; a very long range	Refused, or handled exactly as documented.
Time	00:00; 12:00 AM and PM; 23:59	Stored and shown in the system's time zone.
Dropdown / select	The default; every option; an inactive or out-of-scope option sent by editing the request	Only valid options are saved, checked on the server.
Checkbox / multi-select	None; all; select all, then clear one	Minimum and maximum selections enforced.
File upload	An allowed type; a blocked type renamed (.exe → .jpg); at the size limit; over it; 0 bytes; a very long file name; a duplicate name	Type checked by content, not name; a size-limit message; safe stored names; a working preview.
Photo / camera	Portrait and landscape; a large phone photo; camera permission denied	Correct orientation; compressed if documented; a denied permission is explained.
Location / GPS	Permission denied; poor accuracy; offline	A clear message; accuracy recorded if required.
Password	Below the minimum length; a common password; a mismatched confirmation; pasted	Rules shown and enforced; never displayed, emailed, or logged.

Input test cases

ID	TEST STEPS	EXPECTED RESULT	RESULT
IN-01	Submit a form with errors in several fields.	All errors are shown at once next to their fields; focus moves to the first; entries are kept.	
IN-02	Read each error message.	It says what is wrong and how to fix it; no codes or technical traces.	
IN-03	Bypass the on-screen checks (send the request directly, or edit the page).	The server enforces the same rules.	
IN-04	Use the form with the keyboard only: Tab, Enter, Esc.	A logical order; Enter submits once; Esc closes dialogs.	
IN-05	Use browser autofill and paste.	Fields accept them; validation still runs.	
IN-06	Check the required-field markers.	Every required field is marked; optional fields are not.	
IN-07	Change a field that controls another field's options.	The other field's options and rules update; stale values are cleared.	
IN-08	Leave a long form halfway, then come back.	The draft is restored, or the loss is warned before leaving.	
IN-09	Fill in number fields on a phone.	A numeric keyboard opens.	
IN-10	Open a new form.	Defaults are sensible and documented (for example today, or the user's own <unit>).	

Data outputs and where to verify them

OUTPUT	WHERE TO VERIFY IT	WHAT MUST BE TRUE
Records and statuses	List, detail view, and the database	Values and status match the last action.
<balance>s and ledgers	Summary, ledger, and reports	Opening + in − out = closing.
Audit or activity log	Administrator log screen	Actor, action, <record>, time, and the change.
In-app notifications	Bell, badges, and messages	The right recipients only; counts clear when read.
Emails and SMS	A test inbox	Recipient, subject, branding, and links that work and expire as documented; no secrets.
Generated documents	Invoices, receipts, certificates, letters	Unique, sequential numbers; values match the <record>.
Reports and exports	Downloaded files	See the report cases below.
Integrations	The receiving system or its logs	Complete data; failures are visible and retried.

Output test cases

ID	TEST STEPS	EXPECTED RESULT	RESULT
OU-01	Create, update, approve, and delete a <record>; open the audit log.	One entry for each action with the actor, time, <record>, and change.	
OU-02	As a normal user, try to edit or delete audit log entries.	No such control; the server refuses.	
OU-03	Check who receives each notification.	Only users who may see the <record>; the actor is not notified of their own action unless documented.	
OU-04	Open links in emails sent from the test environment.	They point to the test environment and expire as documented; expired links show a clear message.	
OU-05	Check generated reference numbers after creating and deleting <record>s.	Unique, never reused, and following the documented sequence.	
OU-06	Compare number, date, and unit formats on screen, in files, and in emails.	The same formats and units everywhere.	
OU-07	Make an external service unavailable (with the technical team), then use the feature.	The user is told; no data is lost; it retries or queues as documented.	

Reports and exports

ID	TEST STEPS	EXPECTED RESULT	RESULT
RP-01	Download each report in each format offered (Excel, PDF, CSV).	Each file opens without repair prompts in common apps.	
RP-02	Compare report figures with the screen for the same filters.	Identical.	
RP-03	Check the report's scope as a <unit> user.	Only that <unit>'s and role's data; no hidden columns or sheets with other data.	
RP-04	Download a report for a period with no data.	The file says there is no data, or the option is disabled; never a broken file.	
RP-05	Download the largest realistic period.	Finishes within the agreed time, or runs in the background with progress; no missing rows.	
RP-06	Check the headers and details in the file.	Title, period, filters, generated time, and who it was prepared for.	
RP-07	Check dates, numbers, and units in the cells.	Real dates and numbers that sort correctly, not text; units in the headers.	
RP-08	Check accents and symbols in the data.	Shown correctly; no garbled characters.	
RP-09	Put a value that starts with =, +, − or @ into a text field, then export to Excel or CSV.	Stored as text; no formula runs when the file is opened.	
RP-10	Check the file name.	Names the system, report, and period; no illegal characters.	
RP-11	Print a PDF report in grayscale.	No cut-off columns; page numbers; table headers repeat on each page; still readable.	
RP-12	Download on a slow or dropped connection.	A clear message; a partial file is never presented as complete.	

PART E · CROSS-CUTTING

Shared behavior across the system

Behavior shared by every screen: dialogs and feedback, layout on different screens, accessibility, browsers, speed, unreliable networks, security basics, data integrity, and time and locale.

Dialogs, feedback, and shared tools

ID	TEST STEPS	EXPECTED RESULT	RESULT
XC-01	Start each destructive action (delete, reject, void).	A confirmation dialog names the <record> and uses a danger style; Cancel changes nothing.	
XC-02	Save or submit on each form.	A loading state while working; a success message after; a clear message on failure; the button is disabled while saving.	
XC-03	Watch for native browser alert, confirm, or prompt boxes.	None; the system's own dialogs are used everywhere.	
XC-04	Trigger notifications and badges.	They appear for relevant changes, clear when viewed, and do not fire for the user's own action unless documented.	
XC-05	Send in-app messages, if any.	Delivered to the right person; unread counts are accurate.	
XC-06	Check calendars or schedules, if any.	Events show in the right time zone and only to their audience.	
XC-07	Open help text, tooltips, and empty states.	They explain what to do next; links work.	
XC-08	Open a missing page, cause a server error (with the technical team), and open a page without permission.	A friendly page with a way back; no technical traces.	
XC-09	Compare terms, button labels, date formats, and colors across screens.	The same words and colors mean the same thing everywhere.	
XC-10	Compare the name, logo, and colors on screens, emails, and files.	Consistent everywhere.	

Responsive layout

ID	TEST STEPS	EXPECTED RESULT	RESULT
RL-01	View each main screen at 360, 768, 1024, and 1440 pixels wide.	No sideways page scroll; content reflows; nothing is cut off.	
RL-02	Use the navigation on a phone.	The menu opens and closes; the current section is shown.	
RL-03	View wide tables on a phone.	They scroll sideways inside their own box, or stack; headers stay readable.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
RL-04	Tap buttons and links on a phone.	Targets are about 44 × 44 pixels or larger and not crowded together.	
RL-05	Rotate a phone or tablet while filling in a form.	The layout adapts; no input is lost.	
RL-06	Zoom the browser to 200%.	Content stays readable and usable.	

Accessibility

ID	TEST STEPS	EXPECTED RESULT	RESULT
AX-01	Use the main flows with the keyboard only: Tab, Shift+Tab, Enter, Space, Esc, arrow keys.	Every control can be reached and used; focus never gets trapped.	
AX-02	Watch the focus indicator while tabbing.	Always visible.	
AX-03	Use a screen reader (NVDA, VoiceOver) on the main flows.	Buttons, fields, and images are announced with meaningful names; errors are announced.	
AX-04	Check form labels.	Every input has a visible label; placeholder text is not used as the only label.	
AX-05	Check text and icon contrast with a contrast tool.	At least 4.5:1 for normal text, and 3:1 for large text and icons.	
AX-06	Check images and icons.	Meaningful ones have alternative text; decorative ones are hidden from screen readers.	
AX-07	Turn on the operating system's reduced-motion setting.	Animations are reduced; nothing flashes more than three times a second.	
AX-08	Check page titles and headings.	Each page has a unique title and headings in a logical order.	

Browsers and devices

ID	TEST STEPS	EXPECTED RESULT	RESULT
BC-01	Run a smoke flow (sign in, create, submit, approve, report) in each supported browser.	Works in all of them; versions are recorded.	
BC-02	Open the system in an unsupported or outdated browser.	A clear message, or it still works in a basic way.	
BC-03	Use device features (camera, GPS, file picker) on Android and iOS.	They work, or a denied permission is explained.	
BC-04	Install the app, if it can be installed (PWA or store app), then update it.	It installs, opens, and moves to the new version.	

Performance

ID	TEST STEPS	EXPECTED RESULT	RESULT
PE-01	Load the first screen and each main screen.	Within the agreed time, for example under 3 seconds on office broadband and 6 seconds on mobile data.	
PE-02	Save and submit.	A response within 2 seconds, or a progress indicator.	
PE-03	Open the largest realistic list, search, and report.	Within the agreed time; no timeouts.	
PE-04	Have the agreed number of users work at the same time.	No errors, no noticeable slowdown, no duplicate records.	
PE-05	Leave a screen with live updates open for an hour.	It does not get slower or stop responding.	
PE-06	Throttle the network (browser developer tools, Slow 4G or 3G).	Loading states appear; nothing is submitted twice.	

Network and offline

ID	TEST STEPS	EXPECTED RESULT	RESULT
NW-01	Drop the network while saving, then retry.	An error, or the save is queued as documented; no duplicate or lost <record>.	
NW-02	Work offline, if offline use is supported.	Allowed actions queue with a pending count; other actions say they need a connection.	
NW-03	Reconnect after working offline.	Queued items sync once, in order; conflicts are reported.	
NW-04	Let the session expire while offline, then reconnect.	Sign in again without losing queued work.	

Security basics

Not a penetration test
These are basic checks a QA assessor can run. They do not replace a security review or a penetration test.

ID	TEST STEPS	EXPECTED RESULT	RESULT
SE-01	Open the site with http:// and check the browser's address bar.	Redirected to HTTPS; no mixed-content warnings.	
SE-02	Open the browser developer tools on each main screen.	No secret keys, passwords, or other users' data in the page source, network responses, or browser storage.	
SE-03	Repeat a request with another user's <record> ID.	Refused (for example 403 or 404); no data.	
SE-04	Type `<script>alert(1)</script>` into text fields, then view them everywhere they appear.	Shown as text; never runs.	

ID	TEST STEPS	EXPECTED RESULT	RESULT
SE-05	Upload a file disguised as an allowed type.	Refused; uploaded files are never run or served as web pages.	
SE-06	Send many failed sign-ins or rapid repeated requests.	Slowed, limited, or locked as documented.	
SE-07	Check the session cookies in the developer tools.	Marked Secure, HttpOnly, and SameSite.	
SE-08	Cause errors on purpose (bad input, missing record).	Messages never show stack traces, database errors, or file paths.	
SE-09	Use password-reset and invitation links.	Single use, expiring, and they never reveal whether an email has an account.	
SE-10	Sign out, then replay an earlier request.	Refused; the session has ended on the server.	
SE-11	Review screens and exports that show personal data.	Only the fields each role needs; masked where documented.	
SE-12	Review defect evidence and logs before sharing them.	No passwords, tokens, reset links, or unnecessary personal data.	

Data integrity

ID	TEST STEPS	EXPECTED RESULT	RESULT
DI-01	Two users change the same <balance> at the same time.	The final value is correct; no update is lost.	
DI-02	Cause a server error in the middle of a multi-step save (with the technical team).	All or nothing; no partial records.	
DI-03	Delete a parent record that has child records.	Blocked, cascaded, or kept consistent exactly as documented.	
DI-04	Import a file with a mix of valid and invalid rows, if imports are supported.	Valid rows are imported, or none are, as documented; each invalid row is reported.	
DI-05	Restore a backup into a test environment (with the technical team).	The restored data is complete and consistent.	
DI-06	Check how old and deleted data is kept.	Follows the documented retention policy.	

Time, locale, and language

ID	TEST STEPS	EXPECTED RESULT	RESULT
TL-01	Act just before and just after midnight in the system's time zone.	Each action is dated to the correct day; "today" is correct.	
TL-02	Check month, quarter, and year ends, and 29 February.	Period totals are correct.	
TL-03	Change the computer's time zone, then view dates and times.	Times are shown consistently, as documented.	
TL-04	Check number, money, and date formats.	They match the locale and are consistent.	
TL-05	Switch languages, if the system is multilingual.	No missing or untranslated text; layouts fit longer words.	

APPENDICES

Templates and reference

Fill-in templates that make expected results exact, plus the defect report, the test summary and sign-off, and a short glossary.

A1. Status map

List every status exactly as the system shows it, and which statuses may follow it. Reconciliation case RC-04 checks real histories against this table.

RECORD	STATUS (EXACT WORDING)	MEANING	ALLOWED NEXT STATUSES	WHO CAN CHANGE IT
Request	*Pending*	*Submitted and waiting for review*	*Approved, Rejected, Returned*	*Approver*

A2. Business rules and limits

RULE	VALUE	WHERE IT IS ENFORCED	TEST CASE
Maximum quantity per request	*600*	*Request form and server*	*PF3-03*

A3. Boundary values cheat sheet

INPUT	VALUES TO TRY
Numbers	minimum − 1, minimum, minimum + 1, maximum − 1, maximum, maximum + 1, 0, a negative number
Text length	0, 1, the maximum, maximum + 1 characters
Dates	The first and last day of a period; 23:59 and 00:00; 29 February; 31 December and 1 January
Files	0 bytes; exactly the size limit; the limit + 1 byte
Lists and pages	0 items, 1 item, exactly one page, one page + 1 item
Percentages	0%, 100%, 100.01%, and values that round (99.95%)
Quantities against a <balance>	Exactly the available amount, 1 more, and 1 less

A4. Documentation differences to confirm

When the system and its documents disagree, record it here instead of guessing which is right. The project lead decides which one changes.

#	DOCUMENT AND SECTION	WHAT IT SAYS	WHAT THE SYSTEM DOES	DECISION
1				
2				
3				
4				
5				
6				

A5. Defect report template

FIELD	WHAT TO WRITE
Defect ID / Test case ID	Your defect number and the case it came from (for example PF4-06).
Summary	One line: what is wrong and where.
Severity	Critical, High, Medium, or Low (see Before you test).
Role and account	Role, account email, and <unit> if applicable.
Location	Screen › section › tab, and the address (URL).
Date and time	When it happened, in the system's time zone.
Record identifier	The <record> number or reference shown on screen.
Steps to reproduce	Numbered steps from a known starting state.
Expected result	As written in this guide.
Actual result	What happened, including the exact message.
Evidence	A screenshot or screen recording; the downloaded file if a report is wrong.
Device and browser	For example Windows 11, Chrome 140; Android 15, Chrome.

Never include

Passwords, invitation or reset links, one-time codes, access tokens, or unnecessary personal data in a defect report, screenshot, or recording.

A6. Test summary and sign-off

AREA	CASES	PASS	FAIL	BLOCKED	N/A
Access and roles	20				
Process flow stages 0–8	58				
End-to-end reconciliation	8				
Monitoring	24				
Visualizations	16				
Inputs	10				
Outputs	7				
Reports and exports	12				
Dialogs and shared tools	10				
Responsive layout	6				
Accessibility	8				
Browsers and devices	4				
Performance	6				
Network and offline	4				
Security basics	12				
Data integrity	6				
Time, locale, and language	5				
Total	216				

ROLE	NAME	SIGNATURE	DATE
QA assessor			
QA lead			
Development lead			
Product owner or client representative			

Build tested: _____ Environment: _____ Test period: _____

A7. Glossary

TERM	MEANING
Smoke test	A short check that the build works at all (every role signs in and opens its home screen) before full testing starts.
Regression test	Rerunning earlier cases after a fix or change, to make sure nothing else broke.
UAT	User acceptance testing: the people who will use the system confirm it does what they need.
Severity	How much harm a defect does. Set by QA.
Priority	How soon a defect should be fixed. Set by the project lead; usually, but not always, follows severity.
Blocked	A case that cannot run because something it depends on failed or is missing.
Scope	The data a user may see and change, limited by role and <unit>.
Reconciliation	Checking that the same figure agrees everywhere it appears, and that movements add up to the balance.
Boundary value	A value at, just inside, or just outside a limit, where defects are most common.
Happy path / negative test	A case with valid input that should succeed, and a case with invalid input that should be refused.
Audit trail	The log of who did what and when.
Idempotent	An action that has the same effect whether it runs once or several times, such as a payment that cannot be charged twice.